I/O Wiring and Addressing

DI · DO · AI · AO · Sinking · Sourcing · 4-20 mA

• PLC PROGRAMMING • LESSON 03

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Digital Input Wiring — Sinking and Sourcing

Every real PLC installation begins with wiring field devices to I/O modules. The most common mistake on site — and the one that causes the most confusion — is getting **sinking vs sourcing** wrong. This single concept determines how you connect NPN and PNP sensors to a digital input module, and getting it wrong means the input will never turn on no matter how much you check the program.

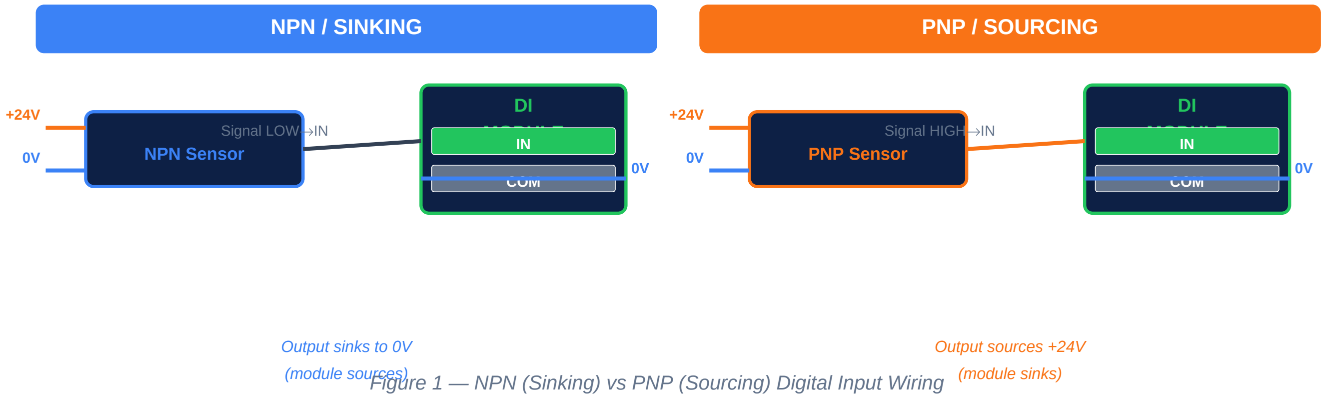

Figure 1 — NPN (Sinking) vs PNP (Sourcing) Digital Input Wiring

Sensor/Output	Signal When Active	Connects To	Module Type Needed
NPN transistor sensor (3-wire, black signal)	Output pulls signal to 0 V (LOW)	Signal → DI Input terminal Sensor 0V = COM	Sinking input module (sources current out of input)
PNP transistor sensor (3-wire, black signal)	Output pushes signal to +24 V (HIGH)	Signal → DI Input terminal Sensor 0V = COM	Sourcing input module (sinks current into input)
Dry contact (switch/relay)	Closes circuit between two terminals	One terminal to DI Input, other to COM No polarity issue	Sinking or sourcing — either works
2-wire proximity sensor (NAMUR type)	Changes impedance when target present	Special NAMUR input card or signal conditioner	NAMUR-compatible input (not standard DI)

Digital Input Module Specifications

Spec	Typical Value	Why It Matters
Input voltage range	10–30 VDC (24 VDC nominal)	Must match your field device supply voltage
Logic "1" threshold	> 11 V (IEC 61131-2 Type 1)	Voltage below this reads as "0" even if device is on
Logic "0" threshold	< 5 V	Voltage above this reads as "1" even if device is off
Input current	5–10 mA per input	Size power supply for total DI current plus outputs
Input filter time	0–16 ms (configurable)	Debounce for mechanical contacts; reduce for fast pulse counting
Galvanic isolation	Usually optocoupled (500–1 500 V)	Protects CPU from field voltage spikes and ground loops

Field wiring tip: Always use shielded cable for DI runs longer than 5 m in industrial environments. Ground the shield at one end only (control panel end) to prevent ground loop current. Never run sensor cables in the same conduit as power cables.

Digital Output and Analog I/O Wiring

Digital outputs (DO) drive actuators — motors, solenoid valves, indicator lamps, and contactor coils. The output type must match the load: **transistor outputs for DC loads**, **relay outputs for high-voltage or AC loads**.

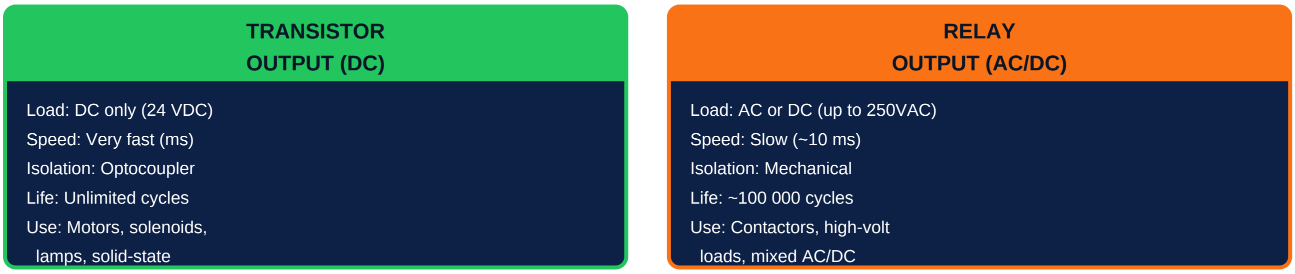

Figure 2 — Digital Output Types: Transistor (DC) vs Relay (AC/DC)

DO Type	Voltage	Max Current	Switching Speed	Typical Load
Transistor (NPN)	5–30 VDC	0.5–2 A per pt	Fast (<1 ms)	Solenoid valves, DC motors, lamps
Transistor (PNP)	5–30 VDC	0.5–2 A per pt	Fast (<1 ms)	Same as NPN; sourcing configuration
Relay (SPDT/SPST)	Up to 250 VAC / 30 VDC	2–8 A per pt	Slow (~10 ms)	Motor starters, AC contactors, heaters
Triac (SSR)	24–240 VAC	0.5–2 A per pt	Very fast	AC loads only; no mechanical wear
Analog Output (AO)	0–10 VDC or 4–20 mA	10–20 mA	N/A (continuous)	VFD speed reference, control valves, positioners

Analog Input Wiring — 4-20 mA and 0-10 V

Analog inputs measure process variables: temperature, pressure, flow, level, speed. The two universal standards are **4-20 mA current loop** (preferred for long cable runs — immune to voltage drop) and **0-10 VDC** (simpler but limited to short runs). 4-20 mA is always preferred in industrial installations.

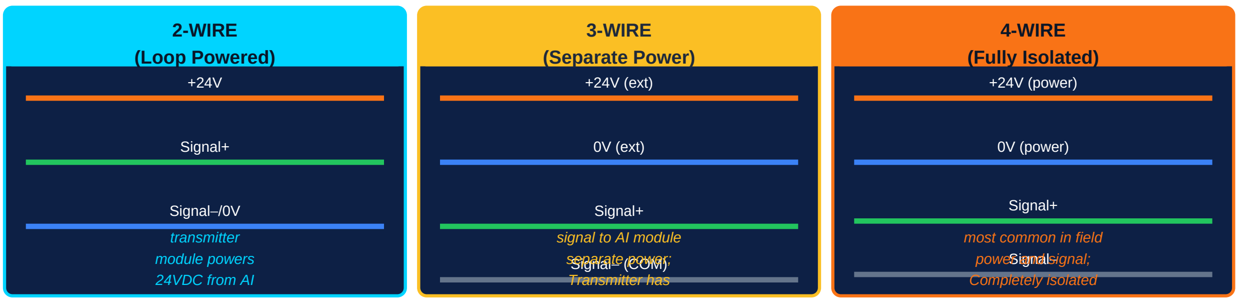

Figure 3 — 4-20 mA Analog Input Wiring: 2-Wire / 3-Wire / 4-Wire

Signal Type	Range	Cable Run	Fault Detection	Typical Use
4-20 mA current loop	4 mA = 0%, 20 mA = 100% (0 mA = wire break)	Up to 1 000 m with 24 AWG	Yes — 0 mA = open circuit fault; < 4 mA = under-range	Pressure, flow, temperature, level transmitters
0-10 VDC voltage signal	0 V = 0%, 10 V = 100%	< 30 m (voltage drop issue)	No — 0 V is a valid signal value	Local proximity sensors, short-range transducers
0-20 mA current loop	0 mA = 0%, 20 mA = 100%	Up to 1 000 m	No — 0 mA cannot distinguish from wire break	Less common; avoid where possible

Pt100 / Pt1000 RTD	Resistance changes with temp (100 Ω at 0°C for Pt100)	Up to 200 m (4-wire)	Open/short circuit detectable on 3/4 wire	Precision temperature measurement; ±0.1°C accuracy
Thermocouple (J/K/T type)	mV signal (1–70 mV range)	Use thermocouple extension wire only	Open circuit detectable	High temperature measurement up to 1 300°C

4-20 mA live-zero advantage: The "live zero" at 4 mA means a broken wire (0 mA) is immediately distinguishable from a 0% process reading (4 mA). This is why 4-20 mA is mandatory for safety-critical measurements — a broken transmitter cable triggers an alarm, not a false "process at zero" reading.

I/O Addressing Across Major PLC Brands

Every I/O point in a PLC has a unique address used in the program to read inputs or write outputs. The address format varies by brand but always encodes the same information: **rack, slot (module), and channel (bit/word position)**. Understanding any brand's addressing scheme comes down to knowing these three elements.

Brand	DI Address Format	DO Address Format	AI Address	Example DI 0
Siemens S7 (TIA Portal)	I[byte].[bit] e.g. I0.0, I0.1 … I0.7	Q[byte].[bit] e.g. Q0.0, Q4.3	IW[word address] e.g. IW64, IW68	I0.0 = first DI bit
Allen-Bradley ControlLogix	Local:[slot]:I.[bit] e.g. Local:1:I.Data.0	Local:[slot]:O.[bit] e.g. Local:2:O.Data.0	Local:[slot]:I.Ch[n] Data	Local:1:I.Data.0
Allen-Bradley MicroLogix/SLC	I:[slot]/[bit] e.g. I:0/0, I:1/3	O:[slot]/[bit] e.g. O:0/0	I:[slot].[word]	I:0/0
Mitsubishi MELSEC iQ-R	X[hex address] e.g. X0, X1 … XF, X10	Y[hex address] e.g. Y0, Y10	D[register] e.g. D0, D10	X0 = first DI
Omron NJ/NX Series	_DI[unit].[bit] or IN_[n]	_DO[unit].[bit] or OUT_[n]	Analog variable tag (tag-based, not address)	IN_00_00
Schneider Modicon M340	%I[word].[bit] e.g. %I0.0	%Q[word].[bit] e.g. %Q0.0	%IW[word] e.g. %IW0	%I0.0

I/O Wiring Best Practices

Practice	Rule	Consequence of Ignoring
Cable segregation	Run sensor/signal cables (< 50 V) in separate conduit/tray from power cables (> 50 V)	Electromagnetic interference causes false inputs, erratic sensor readings
Shield grounding	Ground cable shields at control panel end only; never both ends	Ground loop currents create noise that appears as analog signal drift
Ferrite cores	Fit ferrite beads on analog cables entering the panel	High-frequency RF noise from VFDs/inverters corrupts 4-20 mA signals
Fusing per output	Fuse each DO circuit individually (fast-blow, rated 1.5× load current)	Single fault can take out entire DO module or fire cable
Surge protection	Fit surge protection devices (SPD) on field device cables entering from outside	Lightning strike destroys I/O module and CPU in microseconds
Label every wire	Use heat-shrink labels at both ends; match to I/O list and drawings	Fault-finding takes hours instead of minutes; causes wiring errors on modifications
Unused I/O points	Document and address unused points; fit terminal covers on unused modules	Accidental activation of unused output could start equipment unexpectedly

What comes next — PLC-004: Ladder Logic Fundamentals — contacts, coils, normally-open vs normally-closed logic, the rung structure, and how a PLC executes a ladder program during its scan cycle.